OBESITY



Obesity is linked to several conditions, such as type II diabetes and obstructive sleep apnea, that may increase perioperative risks. Comorbidities may outweigh BMI in assessing these risks. [1-4]

For patients with class 1* obesity (BMI: 30–34.9), perioperative risks are mainly limited to venous thromboembolism. In class 2* obesity (BMI: 35–39.9) and higher, there is a modest increase in risks, including postoperative pulmonary complications, wound infections, longer hospital stays, increased blood loss, longer surgeries, and renal failure. [5-8]

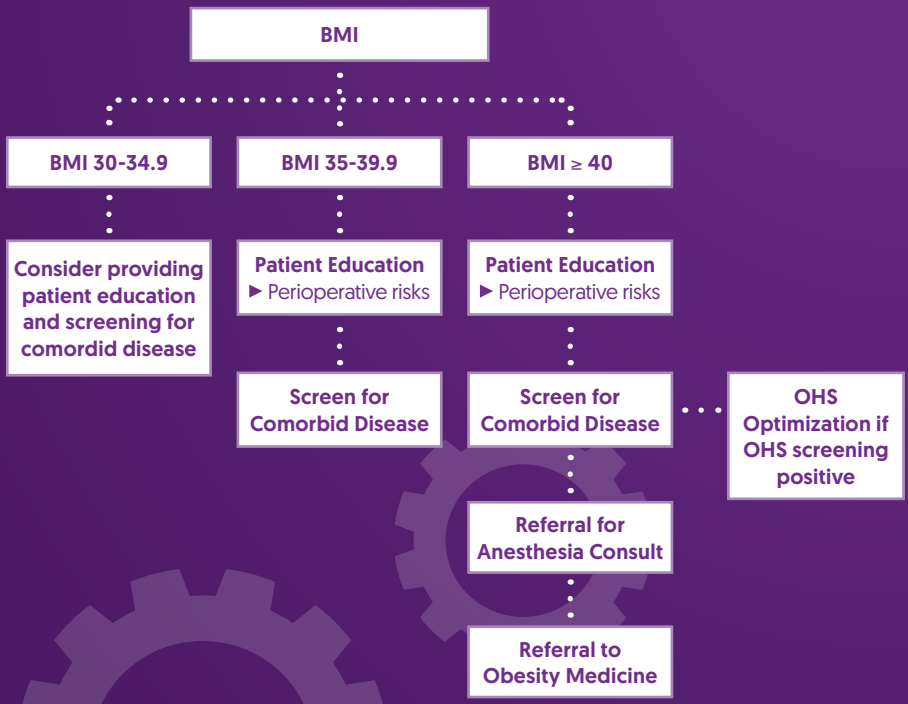
Although high BMI is associated with higher perioperative risks, there is limited research on the effects of preoperative weight loss through diet and limited evidence to suggest delaying surgery based on BMI. [9]

**Classification is based on Caucasian populations. Lower cutoffs have been recommended for other ethnicities.* [10]

Screening Tools

Although BMI is easy to obtain and therefore used to identify and categorize severity of obesity, integrating other indices (e.g., waist to hip ratio, waist circumference, and percent body fat) may improve predictions of metabolic health, comorbid conditions, and perioperative risk stratification. [1-3]

OBESITY ALGORITHM



Prehabilitation and Optimization Recommendations

Patient Education

- Counsel patients on obesity-related perioperative risks, recognizing that stigma and bias can worsen morbidity and mortality
  - Ask permission to measure and discuss weight as it impacts surgical risk "Would it be okay if we discussed your weight today as it impacts your anesthetic and surgical care?"
- Refer to online provider resources for information on communicating and treating obese patients

Screen for Comorbid Disease

- Obstructive Sleep Apnea (OSA)
  Refer to sleep apnea section
- Obesity Hypoventilation Syndrome (OHS)
  See below
- Diabetes
  Refer to glycemic control section
- Cardiovascular disease risk stratification
  Refer to cardiac section
- Kidney disease

Obesity Hypoventilation Syndrome (OHS)

- Initial Screen for OHS
  - Complete Room air ABG if BMI ≥ 50 or BMI ≥ 35 and STOP-Bang ≥ 6 and HC03 ≥ 27
  - For patients with BMI < 50, a serum HCO3 < 27 effectively rules out OHS
- Room air PaCO2 ≥ 45 mmHg suggests OHS, but other potential causes of hypercapnia need to be ruled out for a definitive diagnosis
- If OHS suspected:
  - Referral for Anesthesia consult
  - Referral to Sleep Medicine
  - Preoperative investigations
    - Chest X-ray
    - Echocardiogram
    - Spirometry

Referral for Anesthesia Consult

- To allow for assessment of and planning for the perioperative management of comorbid conditions (e.g.,OSA, OHS) and other factors such as airway management, vascular access, and patient positioning [11]

Consider Referral to Obesity Medicine

- Recommended for patients with significant comorbid disease
- Refer to online provider resources for obesity medicine providers